

WEEK 12

PRACTICE SESSION 1

Coding: Attempt review | REC-CIS - Personal - Microsoft Edge

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REC-CIS

Question 1
Correct
Marked out of 1.00
Flag question

A binary number is a combination of 1s and 0s. Its n^{th} least significant digit is the n^{th} digit starting from the right starting with 1. Given a decimal number, convert it to binary and determine the value of the 4^{th} least significant digit.

Example

number = 23

- Convert the decimal number 23 to binary number: $23^{10} = 2^4 + 2^2 + 2^1 + 2^0 = (10111)_2$.
- The value of the 4^{th} index from the right in the binary representation is 0.

Function Description

Complete the function fourthBit in the editor below.

fourthBit has the following parameter(s):

int number: a decimal integer

Returns:

int: an integer 0 or 1 matching the 4th least significant digit in the binary representation of number.

Constraints

$0 \leq \text{number} < 2^{31}$

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```
5  * The function accepts INTEGER number as parameter.
6  */
7
8  int fourthBit(int number)
9  {
10     int binary[32];
11
12     int i=0;
13
14     while(number>0)
15     {
16         binary[i]=number%2;
17         number/=2;
18         i++;
19     }
20     if(i>4)
21     {
22         return binary[3];
23     }
24     else
25         return 0;
26
27 }
28
```

Test	Expected	Got	
✓ printf("%d", fourthBit(32))	0	0	✓
✓ printf("%d", fourthBit(77))	1	1	✓

Passed all tests! ✓

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Question 2
Correct
Marked out of 1.00
Flag question

Determine the factors of a number (i.e., all positive integer values that evenly divide into a number) and then return the p^{th} element of the list, sorted ascending. If there is no p^{th} element, return 0.

Example

$n = 20$
 $p = 3$

The factors of 20 in ascending order are {1, 2, 4, 5, 10, 20}. Using 1-based indexing, if $p = 3$, then 4 is returned. If $p > 6$, 0 would be returned.

Function Description

Complete the function `pthFactor` in the editor below.

`pthFactor` has the following parameter(s):

- `int n`: the integer whose factors are to be found
- `int p`: the index of the factor to be returned

Returns:

`int`: the long integer value of the p^{th} integer factor of n or, if there is no factor at that index, then 0 is returned.

Constraints

Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share

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```
8 */
9
10 long pthFactor(long n, long p)
11 {
12     int count=0;
13
14     for(long i=1;i<=n;i++)
15     {
16         if(n%i==0)
17         {
18             count++;
19             if(count==p)
20             {
21                 return i;
22             }
23         }
24     }
25     return 0;
26 }
27
28
29
```

Test	Expected	Got	
✓ printf("%ld", pthFactor(10, 3))	5	5	✓
✓ printf("%ld", pthFactor(10, 5))	0	0	✓
✓ printf("%ld", pthFactor(1, 1))	1	1	✓

Passed all tests! ✓

PRACTICE SESSION 2

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REC-CIS

Correct
Marked out of 1.00
Flag question

You are a bank account hacker. Initially you have 1 rupee in your account, and you want exactly **N** rupees in your account. You wrote two hacks, first hack can multiply the amount of money you own by 10, while the second can multiply it by 20. These hacks can be used any number of time. Can you achieve the desired amount **N** using these hacks.

Constraints:

$1 \leq T \leq 100$
 $1 \leq N \leq 10^{12}$

Input

The test case contains a single integer N.

Output

For each test case, print a single line containing the string "1" if you can make exactly N rupees or "0" otherwise.

SAMPLE INPUT

1

SAMPLE OUTPUT

1

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```
1 /*  
2  * Complete the 'myFunc' function below.  
3  *  
4  * The function is expected to return an INTEGER.  
5  * The function accepts INTEGER n as parameter.  
6  */  
7  
8 int myFunc(int n)  
9 {  
10     return n==1 || n%10==0;  
11 }  
12  
13
```

	Test	Expected	Got	
✓	printf("%d", myFunc(1))	1	1	✓
✓	printf("%d", myFunc(2))	0	0	✓
✓	printf("%d", myFunc(10))	1	1	✓
✓	printf("%d", myFunc(25))	0	0	✓
✓	printf("%d", myFunc(200))	1	1	✓

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1.00
Flag question

For example, if $X = 13$ and $N = 2$, we have to find all combinations of unique squares adding up to 13 . The only solution is $2^2 + 3^2$.

Function Description

Complete the powerSum function in the editor below. It should return an integer that represents the number of possible combinations.

powerSum has the following parameter(s):

- X : the integer to sum to
- N : the integer power to raise numbers to

Input Format

The first line contains an integer X .
The second line contains an integer N .

Constraints

- $1 \leq X \leq 1000$
- $2 \leq N \leq 10$

Output Format

Output a single integer, the number of possible combinations calculated.

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REC-CIS

reset answer

```
1. /*
2.  * Complete the 'powerSum' function below.
3.  *
4.  * The function is expected to return an INTEGER.
5.  * The function accepts following parameters:
6.  * 1. INTEGER x
7.  * 2. INTEGER n
8.  */
9. #include <math.h>
10. int powerSum(int x, int m, int n)
11. {
12.     int p=pow(m,n);
13.     if (p==x)
14.         return 1;
15.     if (p>x)
16.         return 0;
17.     return powerSum(x-p,m+1,n)+powerSum(x,m+1,n);
18.
19.
20. }
```

Test	Expected	Got
✓ printf("%d", powerSum(10, 1, 2))	1	1 ✓

Passed all tests! ✓

Finish review